

Shubham Acharya

Bengaluru, Karnataka | +91-6350300258 | shubhamacharya258@gmail.com

github.com/shubhamacharya02 | linkedin.com/in/shubham-acharya

Professional Summary

AI Engineer / Backend Developer with hands-on experience building production-oriented Generative AI applications and workflow automation systems using Python, FastAPI, LangChain, LangGraph, RAG, and LLM APIs. Experienced in designing multi-agent systems, dynamic AI workflow engines, modular plugin architectures, vector search pipelines, and scalable REST APIs. Proficient in PostgreSQL, Docker, Google Cloud, Supabase, OAuth/JWT authentication, and modern LLM ecosystems.

Technical Skills

Languages: Python, SQL

AI & Generative AI: LangChain, LangGraph, RAG, AI Agents, Multi-Agent Systems, Prompt Engineering, Tool Calling, Vector Search, Embeddings, Transformer Architecture, NLP, Hugging Face, MCP

LLM Evaluation: LangSmith, DeepEval, LLM Evaluation, RAG Evaluation, Evaluation Metrics

Machine Learning: Machine Learning

Backend: FastAPI, REST APIs, SQLAlchemy, Pydantic, OAuth 2.0, JWT

Data & ML Libraries: Pandas, NumPy, scikit learn

Databases & Storage: PostgreSQL, MySQL, MongoDB, ChromaDB, Supabase

LLMs & AI Providers: Google Gemini, Vertex AI, GPT, Claude, Groq, Llama

Cloud & DevOps: Docker, Docker Compose, Google Cloud Platform, Cloud Run, Vercel, Render

Tools: Git, GitHub

Projects

NexusAI — AI Workflow Automation & Research Platform

Python • FastAPI • LangGraph • LangChain • Gemini • PostgreSQL • Docker

Built a visual, node-based AI workflow platform enabling users to create and execute custom multi-step research and knowledge pipelines. Designed a modular plugin architecture using metadata, Pydantic schemas, and independent execution modules, allowing new AI nodes to be added without modifying the core execution engine. Developed a dynamic LangGraph execution engine that converts frontend workflow graphs into executable StateGraph workflows and resolves data dependencies between connected AI nodes. Implemented server-driven node discovery that dynamically scans backend plugins and exposes metadata and JSON Schemas for automatic frontend tool and configuration-panel generation.

PrepMate AI — AI-Powered Interview Preparation Platform

LangGraph • LangChain • FastAPI • Vertex AI • PostgreSQL • Docker

Built an AI-powered interview preparation platform featuring resume analysis, ATS scoring, AI mock interviews, job search, and personalized career roadmaps. Designed LangGraph-based workflows for resume evaluation, interview question generation, candidate assessment, and career planning. Developed and deployed secure FastAPI services using Google OAuth, JWT authentication, PostgreSQL, Supabase, Docker, and Google Cloud.

Medical AI Assistant — Retrieval-Augmented AI System

LangGraph • LangChain • FastAPI • ChromaDB

Developed a RAG-based medical information assistant with intelligent tool routing for evidence-backed responses. Integrated PubMed, OpenFDA, Tavily Search, and Wikipedia for multi-source information retrieval. Built a PDF ingestion pipeline using semantic chunking, embeddings, ChromaDB vector storage, and citation-aware response generation.

Experience

Digital Computer Instructor

Dec 2025 – Mar 2026

BR Birla Group of Schools, Jodhpur

Conducted hands-on robotics and programming sessions covering Python, Arduino Uno/Nano, Raspberry Pi, sensors, servo motors, and basic electronics. Guided students in building robotics projects, circuit connections, and hardware control through microcontroller programming.

Computer Instructor

Jul 2025 – Nov 2025

Parashar International School, Jodhpur

Taught Python and robotics through practical projects using Arduino, Raspberry Pi, sensors, servo motors, and electronic components. Guided students in circuit diagrams, hardware integration, programming, and project-based problem solving.

Education

Bachelor of Computer Applications (BCA)

2022 – 2025

Aishwarya College, Jodhpur

CGPA: 7.0